

Qatar-1b

R-0616

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_240910 · created 2026-07-25T10:36:38+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460563.7541 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.013
Transit depth (Rp/Rs)^2	2.69 +/- 0.44 [%]
Orbital Inclination (inc)	83.1 +/- 0.6
Airmass coefficient 1 (a1)	1.097 +/- 0.01
Airmass coefficient 2 (a2)	-0.07 +/- 0.017
Scatter in the residuals of the lightcurve fit is	0.93 %
Best Comparison Star	None
Optimal Aperture	2.92
Optimal Annulus	11.65
Transit Duration (day)	0.0645 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.013 vs 0.1463 (deviation 0.95σ)
Duration fit vs published	0.0645 d vs 0.0692 d (deviation 0.74σ)
Mid-time O–C	-5.1 min
Depth SNR	8.8
Residual scatter	0.93 %

Light curve

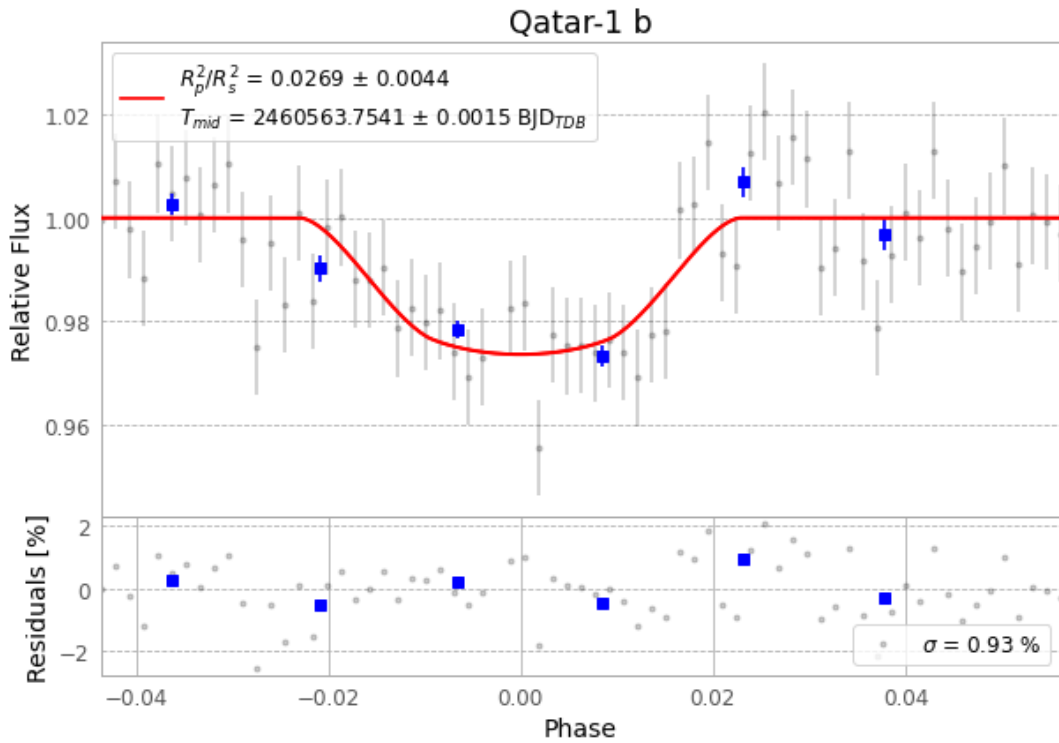


Figure 1 — FinalLightCurve_Qatar-1 b_2024-09-09.png (SHA-256 d6a44501f722cdfb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [387, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 423ff0f4e72f3574...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

69 FITS frames (45 MB), Qatar-1240910043316.FITS → Qatar-1240910075715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-240910.log (306f9395c11e...), FinalLightCurve_Qatar-1 b_2024-09-09.png (d6a44501f722...), FinalLightCurve_Qatar-1 b_2024-09-09.csv (5aebdeea5351...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 10:36Z.